Wheel Classes in Kontsevich Graph Complex and Merkulov's Low-Valence Conjecture

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Abstract

We show that the wheel classes in the Kontsevich graph complex GC_d admit representatives supported on graphs with only 3- and 4-valent vertices. This verifies that Merkulov's low-valence conjecture holds for the wheel classes. More precisely, for every $m \geq 2$, we prove that the wheel graph W_{2m+1} is homologous to an explicit linear combination of 2^{m-2} graphs, each having only 3- and 4-valent vertices.

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1 Introduction

A conjecture of Merkulov predicts that every homology class in the Kontsevich graph complex GC_d should admit a representative supported on graphs whose vertices all have valence 3 or 4.

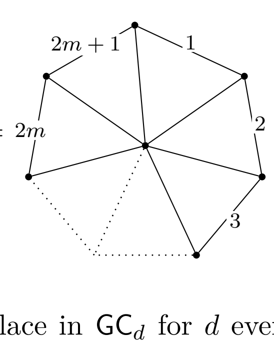
Conjecture 1.1 (Merkulov)**.** *Let $\mathsf{GC}_d^{\mathrm{Me}}$ be the maximal subcomplex of GC_d spanned by linear combinations of graphs all of whose vertices have valence 3 or 4. Then the inclusion*

$$i\colon \mathsf{GC}_d^{\mathrm{Me}} \hookrightarrow \mathsf{GC}_d$$

is a quasi-isomorphism.

This conjecture has been checked by Brun and Willwacher up to genus 10 in all cohomological degrees [1].

In this paper we show that the homology classes in the even graph complex represented by the wheel graphs

$W_{2m+1} =$ 

are consistent with Merkulov's conjecture.

These classes occupy a distinguished place in GC_d for d even. By Willwacher's identification of the degree-zero cohomology of GC_2 with the Grothendieck–Teichmüller Lie algebra [3], they correspond to the conjectural generators of $\mathfrak{grt}_1$; see also [2].

The wheel graphs W_{2m+1} are obviously not contained in $\mathsf{GC}_d^{\mathrm{Me}}$ for $m > 1$, since they contain one $2m+1$-valent hub vertex. Our main theorem shows that the homology classes associated with the wheel graphs nevertheless admit representatives in $\mathsf{GC}_d^{\mathrm{Me}}$.

Theorem. *For d even there exists an element $U_{2m+1} \in \mathsf{GC}_d$ such that*

$$W_{2m+1} - d(U_{2m+1}) \in \mathsf{GC}_d^{\mathrm{Me}}.$$

An explicit version, with a formula for U_{2m+1}, is stated and proved later as Theorem 5.1.

2 Conventions

We shall assume familiarity with Kontsevich's graph complex in this paper. To fix conventions, we will consider GC_d to consist of connected graphs in which every vertex is at least 3-valent.

We will use vertex and edge labels starting from 0. We will exclusively consider the case that d is even. In this case, vertex labels and edge directions do not matter, and edge labels matter only through the sign orientation.

The differential d is the edge-contraction differential. If a graph has m edges, then the contraction of the edge labelled i carries the sign

$$(-1)^{i+m}.$$

After contracting the edge labelled i, we relabel the remaining edges by preserving their relative order: labels smaller than i remain unchanged, while each label greater than i is decreased by 1.

3 Two Families of Graphs

The proof is organized around two explicit families of graphs. The family $V_N(S)$ describes the low-valence support that remains after eliminating the higher-valence part of the wheel class, while the family $U_N(S)$ provides the auxiliary chains whose differentials perform this elimination. The construction of $V_N(S)$ and $U_N(S)$ may be summarized as follows.

Construction of $V_N(S)$ and $U_N(S)$.

1. Input: an odd integer N and a binary sequence $S = (s_1, \ldots, s_k)$ of choices, with $s_i \in \{\mathrm{left}, \mathrm{right}\}$.

2. Initialize the graph with edges

$$(0,1),\ (0,2),\ (0,3),\ (1,2),\ (2,3).$$

   Denote vertex 1 by L, vertex 2 by C and vertex 3 by R.

3. For $i = 1, \ldots, k$:

   (a) Add a new vertex attached to the current centre vertex C, and denote the new vertex by `new_C`. The new edge $(\texttt{C}, \texttt{new\_C})$ gets the label $4i+1$.

   (b) Add an edge $(\texttt{new\_C}, R)$ (if $s_i = \mathrm{left}$, we instead add the edge $(\texttt{new\_C}, L)$). This edge gets the label $4i+2$.

   (c) Add a second new vertex denoted `new_R` (or `new_L`). Add an edge $(\texttt{R}, \texttt{new\_R})$ (or $(\texttt{L}, \texttt{new\_L})$). This new edge gets the label $4i+3$.

(d) Add an edge (`new_C`, `new_R`) (or (`new_C`, `new_L`)). This edge gets the label $4i + 4$.

(e) Denote the vertices `new_C` and `new_R` (or `new_L`) by C and R (or L) respectively.

After these k steps, the graph has $2k + 4$ vertices and $4k + 5$ edges.

4. (Only for $U_N(S)$) Add one further centre vertex `new_C` and an edge (`C`, `new_C`). This edge gets the label $4k + 5$. Denote the new vertex C.

5. Now, for both $V_N(S)$ and $U_N(S)$, let m denote the number of remaining vertices to be added; explicitly, $m = N - 2k - 3$. Let n denote the current largest vertex label. If $m = 0$, add the final edge (L, R), with edge label $2N - 1$ (this case will only be considered for $V_N(S)$). Otherwise, add the remaining vertices and edges as an arc of triangles as follows: Let us denote the least unassigned edge label $e = 4k + 5$ for $V_N(S)$ and $e = 4k + 6$ for $U_N(S)$.

   (a) Add $(L, n + 1)$ with edge label e, and add $(n + 1, C)$ with edge label $e + 1$.

   (b) For $i = 1, \ldots, m - 1$: Add $(n + i, n + i + 1)$ with edge label $e + 2i$, and add $(n + i + 1, C)$ with edge label $e + 2i + 1$.

   (c) Add $(n + m, R)$ with edge label $2N - 1$ for $V_N(S)$ and $2N$ for $U_N(S)$.

The construction is illustrated in the following two examples, shown in Figures 1 and 2.

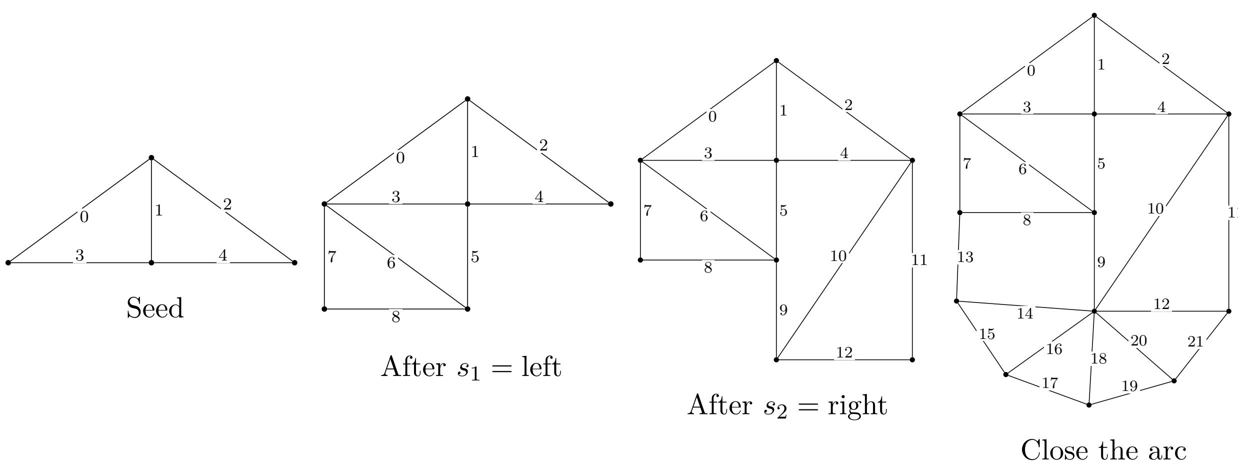


Figure 1: Step-by-step construction of $V_{11}(\text{left}, \text{right})$.

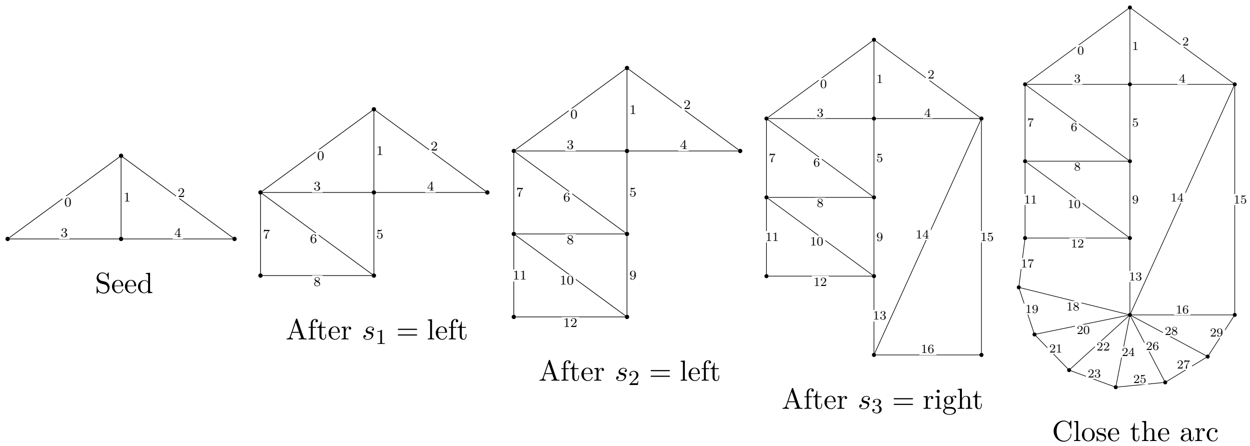


Figure 2: Step-by-step construction of $V_{15}(\text{left}, \text{left}, \text{right})$.

We now record some simple structural properties of the graphs constructed above.

Proposition 3.1. *Let S^{op} be the binary sequence obtained from S by interchanging* left *and* right *in every entry. Then the following statements hold.*

1. *The graphs $V_N(S)$ and $V_N(S^{\text{op}})$ are isomorphic. Likewise, $U_N(S)$ and $U_N(S^{\text{op}})$ are isomorphic. If N is odd, then the isomorphism has an even edge permutation.*

2. *The graph $V_N(\emptyset)$ is the wheel graph W_N.*

3. *For a binary sequence S of length $(N-3)/2$, every vertex of $V_N(S)$ has valence at most 4.*

Proof. For item 1, the isomorphism is given by reversing the order in which we add the edges in Steps 2 and 5. Keep in mind that a permutation reversing the order of n items is even if $n \equiv 0, 1 \pmod 4$ and odd otherwise. In Step 2, we add 5 edges, which reverses with two vertex transpositions. In Step 5, we add $2N - 4k - 5$ edges. For $N = 2m + 1$, we have

$$2N - 4k - 5 = 4m - 4k - 3 \equiv 1 \pmod 4,$$

so this reversal is even.

For $V_N(\emptyset)$, we skip from Step 2 directly to Step 5. This results in a wheel graph where vertex 2 is the centre hub vertex.

Finally, we note that for any $V_N(S)$, the only vertex which may be more than 4-valent is the vertex denoted C in Step 5. For a sequence of length $(N-3)/2$, we do not have any extra edges to attach to this vertex.

□

Examples of V_N and U_N

We now illustrate some examples of graphs in the families $V_N(S)$ and $U_N(S)$ in Figure 3.

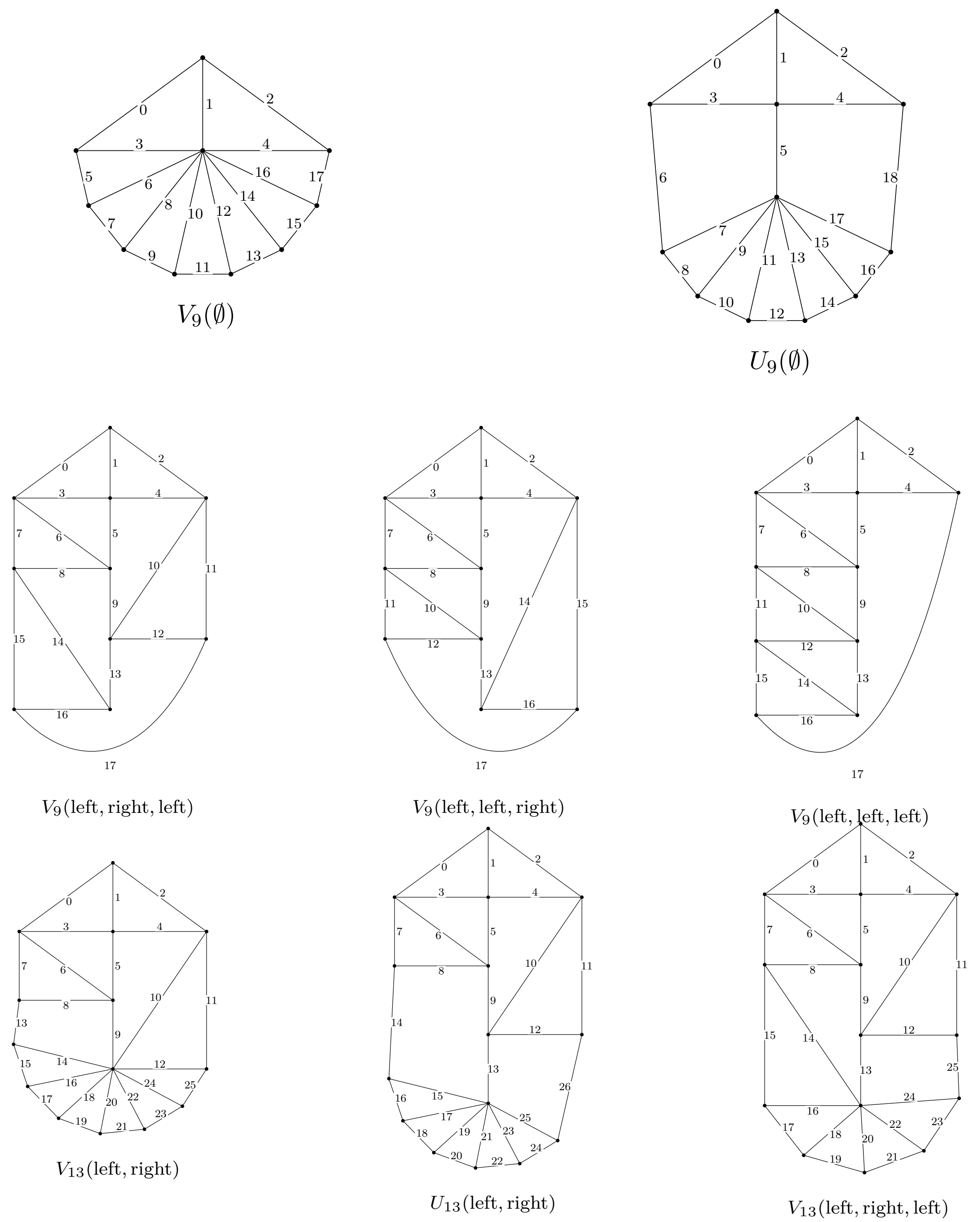


Figure 3: Additional examples in the families $V_N(S)$ and $U_N(S)$.

The families $V_N(S)$ and $U_N(S)$ constructed in this way will be used to produce explicit low-valence representatives for the wheel classes.

4 Differential of $U_N(S)$

Let us now consider the edge contraction differential on $U_N(S)$.

The aim of this section is to prove the following lemma, which forms the core combinatorial step in the proof of the main theorem.

Lemma 4.1. *Let $\mathcal{S}_k$ denote the set of all binary sequences of choices* $\{\text{left}, \text{right}\}$ *of length k. Then, for N odd and $k \leq (N-5)/2$, we have*

$$d\left(\sum_{S\in\mathcal{S}_k} U_N(S)\right) = \sum_{T\in\mathcal{S}_{k+1}} V_N(T) - \sum_{S\in\mathcal{S}_k} V_N(S).$$

Proof. For each $U_N(S)$ there are only a handful of edges that may be contracted without creating a double edge, which would kill the resulting graph in the even graph complex.

The edges that may be contracted are the following.

1. The edge labelled $4k+5$, created in Step 4. The resulting graph is identical to $V_N(S)$. The sign of this contraction is
$$(-1)^{4k+5+2N} = -1.$$

2. The edge labelled $4k+6$, created in Step 5a. This creates a graph that is identical to $V_N(S, \text{left})$. The sign of this contraction is $+1$.

3. The edge labelled $2N$, created in Step 5c. This creates a graph that is isomorphic to $V_N(S, \text{right})$. The sign of this contraction is $+1$.

   The isomorphism is given by swapping the edges labelled $2N-1$ and $2N-3$ and by the three insertions
$$\begin{aligned} 2N-3 &\mapsto 4(k+1)+1, \\ 2N-2 &\mapsto 4(k+1)+2, \\ 2N-1 &\mapsto 4(k+1)+3. \end{aligned}$$

   All these four permutations carry an odd sign, so the composed sign is even.

4. Finally, some edges $4i+1$, for $i = 2, \ldots, k$, created in Step (a), may be contracted without creating a double edge. Let us denote such a graph by $\overline{U}^i_N(S)$.

Consider
$$\overline{U}^i_N(s_1, \ldots, s_{i-1}, s_i, \ldots, s_k)$$
for a sequence $S = (s_1, \ldots, s_k)$. If $s_i = s_{i-1}$, then $\overline{U}^i_N(S)$ vanishes because it contains a double edge.

Otherwise, the graphs
$$\overline{U}^i_N(s_1, \ldots, s_{i-2}, \text{left}, \text{right}, \ldots, s_k)$$
and
$$\overline{U}^i_N(s_1, \ldots, s_{i-2}, \text{right}, \text{left}, \ldots, s_k)$$
are isomorphic, where the isomorphism is given by the three edge transpositions coming from (b), (c), and (d) of Step 3 in the construction list above. Therefore,

$$\overline{U}^i_N(s_1, \ldots, s_{i-1}, s_i, \ldots, s_k) = -\overline{U}^i_N(s_1, \ldots, s_{i-2}, s_i, s_{i-1}, s_{i+1}, \ldots, s_k).$$

Hence these terms cancel pairwise in the sum over all sequences. It follows that

$$\begin{aligned}\sum_{S\in\mathcal{S}_k} d(U_N(S)) &= \sum_{S\in\mathcal{S}_k}\Big(V_N(S,\text{left}) + V_N(S,\text{right}) - V_N(S)\Big)\\ &= \sum_{T\in\mathcal{S}_{k+1}} V_N(T) - \sum_{S\in\mathcal{S}_k} V_N(S).\end{aligned}$$

□

5 Proof of the Main Theorem

We now state the explicit form of the main theorem, in which U_{2m+1} is defined.

Theorem 5.1. *For $k \geq 1$, let $\mathcal{S}_k^{\text{left}} \subset \mathcal{S}_k$ denote the subset of binary sequences of length k whose first entry is* left*. For each $m \geq 2$, define*

$$U_{2m+1}^{(0)} := U_{2m+1}(\emptyset),$$

and

$$U_{2m+1}^{(k)} := 2\sum_{S\in\mathcal{S}_k^{\text{left}}} U_{2m+1}(S), \qquad k = 1,\dots,m-2,$$

and

$$U_{2m+1} := \sum_{k=0}^{m-2} U_{2m+1}^{(k)} = U_{2m+1}(\emptyset) + 2\sum_{k=1}^{m-2}\sum_{S\in\mathcal{S}_k^{\text{left}}} U_{2m+1}(S).$$

Then

$$W_{2m+1} + d(U_{2m+1}) = 2\sum_{S\in\mathcal{S}_{m-1}^{\text{left}}} V_{2m+1}(S).$$

By Proposition 3.1, item 3, every graph on the right-hand side has only 3- and 4-valent vertices.

Proof. By Proposition 3.1, item 1,

$$U_{2m+1} = \sum_{k=0}^{m-2}\sum_{S\in\mathcal{S}_k} U_{2m+1}(S).$$

By Lemma 4.1,

$$d(U_{2m+1}) = \sum_{k=0}^{m-2}\left(\sum_{T\in\mathcal{S}_{k+1}} V_{2m+1}(T) - \sum_{S\in\mathcal{S}_k} V_{2m+1}(S)\right) = \sum_{S\in\mathcal{S}_{m-1}} V_{2m+1}(S) - V_{2m+1}(\emptyset).$$

By Proposition 3.1, item 2, the graph $V_{2m+1}(\emptyset)$ is the wheel graph W_{2m+1}. Hence

$$W_{2m+1} + d(U_{2m+1}) = \sum_{S\in\mathcal{S}_{m-1}} V_{2m+1}(S).$$

Again by Proposition 3.1, item 1,

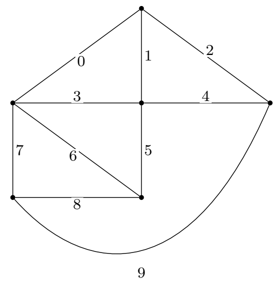
$$\sum_{S\in\mathcal{S}_{m-1}} V_{2m+1}(S) = 2\sum_{S\in\mathcal{S}_{m-1}^{\text{left}}} V_{2m+1}(S).$$

This proves the theorem. □

6 Examples

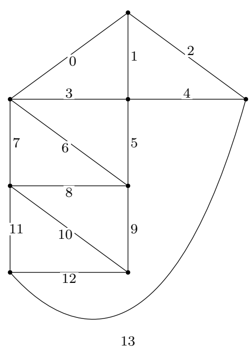
Finally we present visualisations of the explicit 3- and 4-valent representatives of the homology classes W_5, W_7, W_9, and W_{11} provided by Theorem 5.1:

$W_5 \simeq$



$V_5(\text{left})$

$W_7 \simeq$



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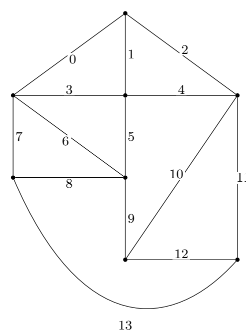
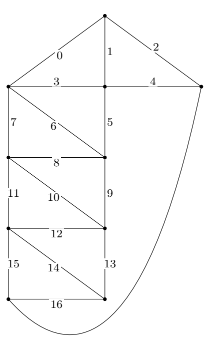


$V_7(LL)$ $V_7(LR)$

$W_9 \simeq$



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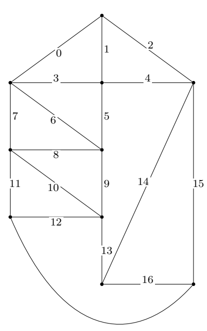

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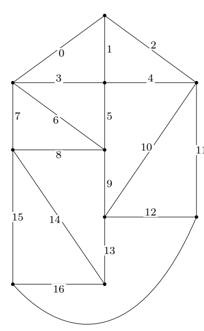

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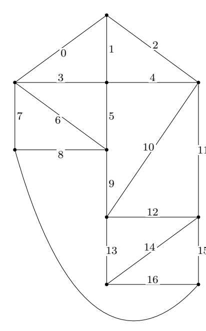
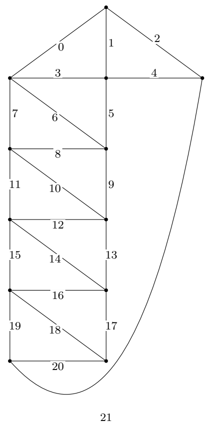
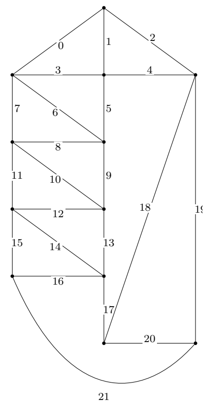
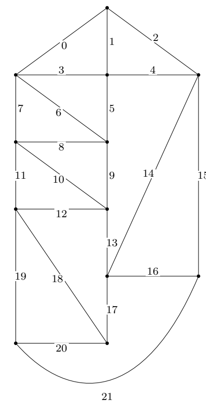
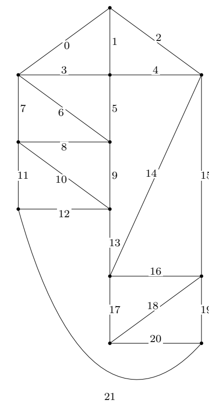
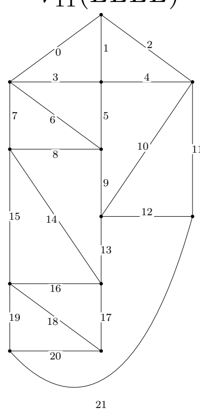
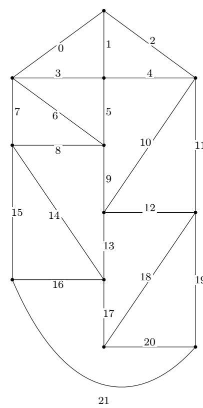
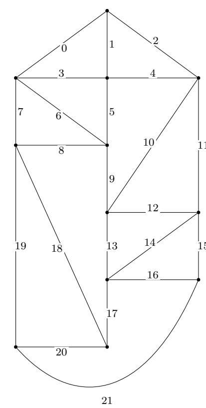
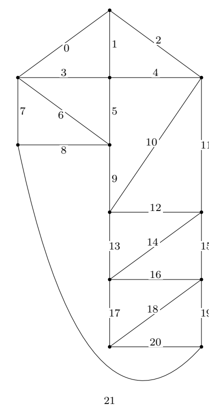


$V_9(LLL)$ $V_9(LLR)$ $V_9(LRL)$ $V_9(LRR)$

$W_{11} \simeq$  $V_{11}(LLLL)$ $+$  $V_{11}(LLLR)$ $+$  $V_{11}(LLRL)$ $+$  $V_{11}(LLRR)$

$+$  $V_{11}(LRLL)$ $+$  $V_{11}(LRLR)$ $+$  $V_{11}(LRRL)$ $+$  $V_{11}(LRRR)$

References

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[2] S. Merkulov, *Grothendieck–Teichmüller group, operads and graph complexes: a survey*, in *Integrability, Quantization, and Geometry: II. Quantum Theories and Algebraic Geometry*, Proc. Sympos. Pure Math. **103.2**, Amer. Math. Soc., 2021, pp. 383–445.

[3] T. Willwacher, *M. Kontsevich's graph complex and the Grothendieck–Teichmüller Lie algebra*, *Inventiones Mathematicae* **200** (2015), no. 3, 671–760. doi:10.1007/s00222-014-0528-x.